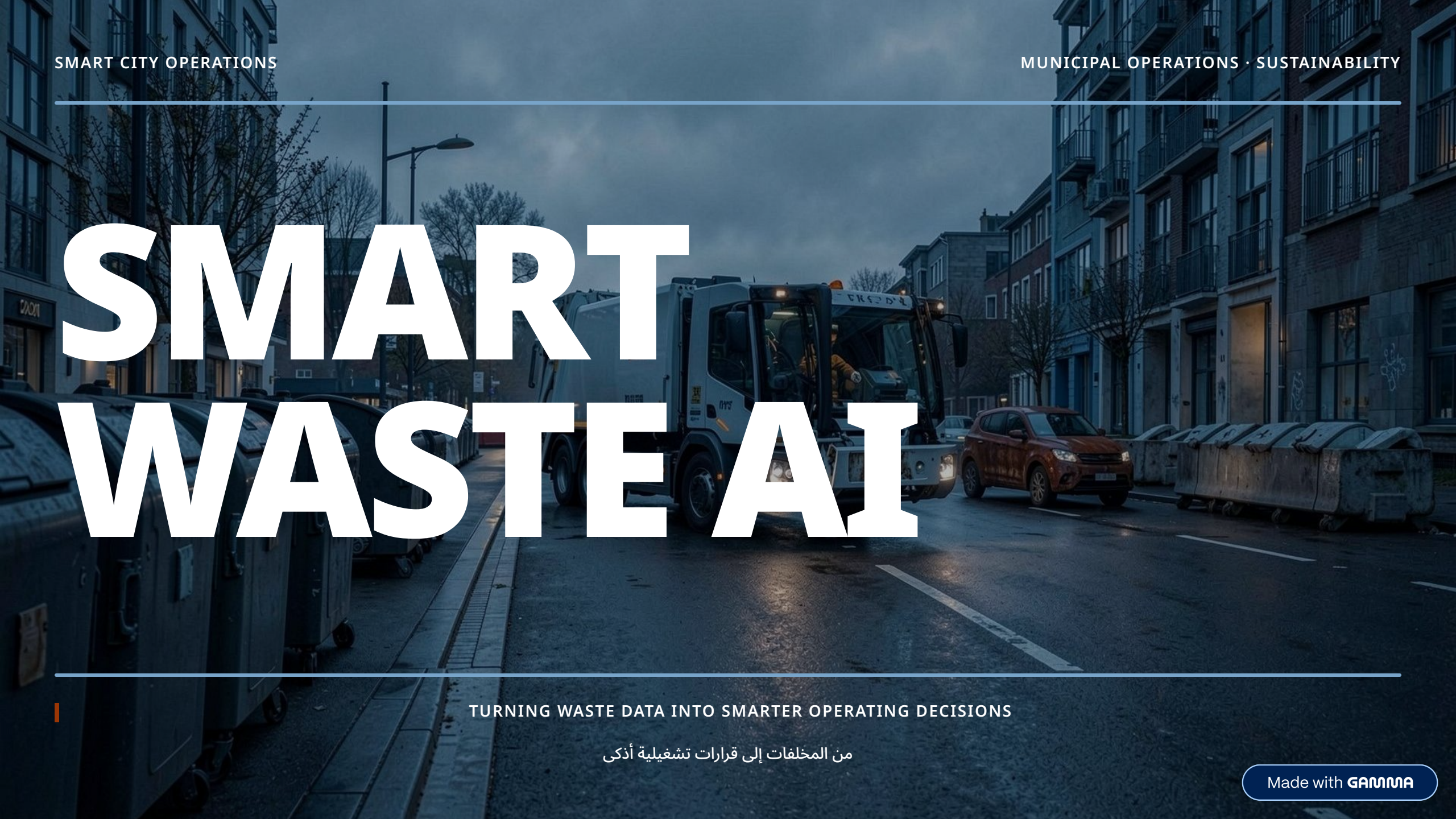


SMART WASTE AI

TURNING WASTE DATA INTO SMARTER OPERATING DECISIONS

من المخلفات إلى قرارات تشغيلية أذكى

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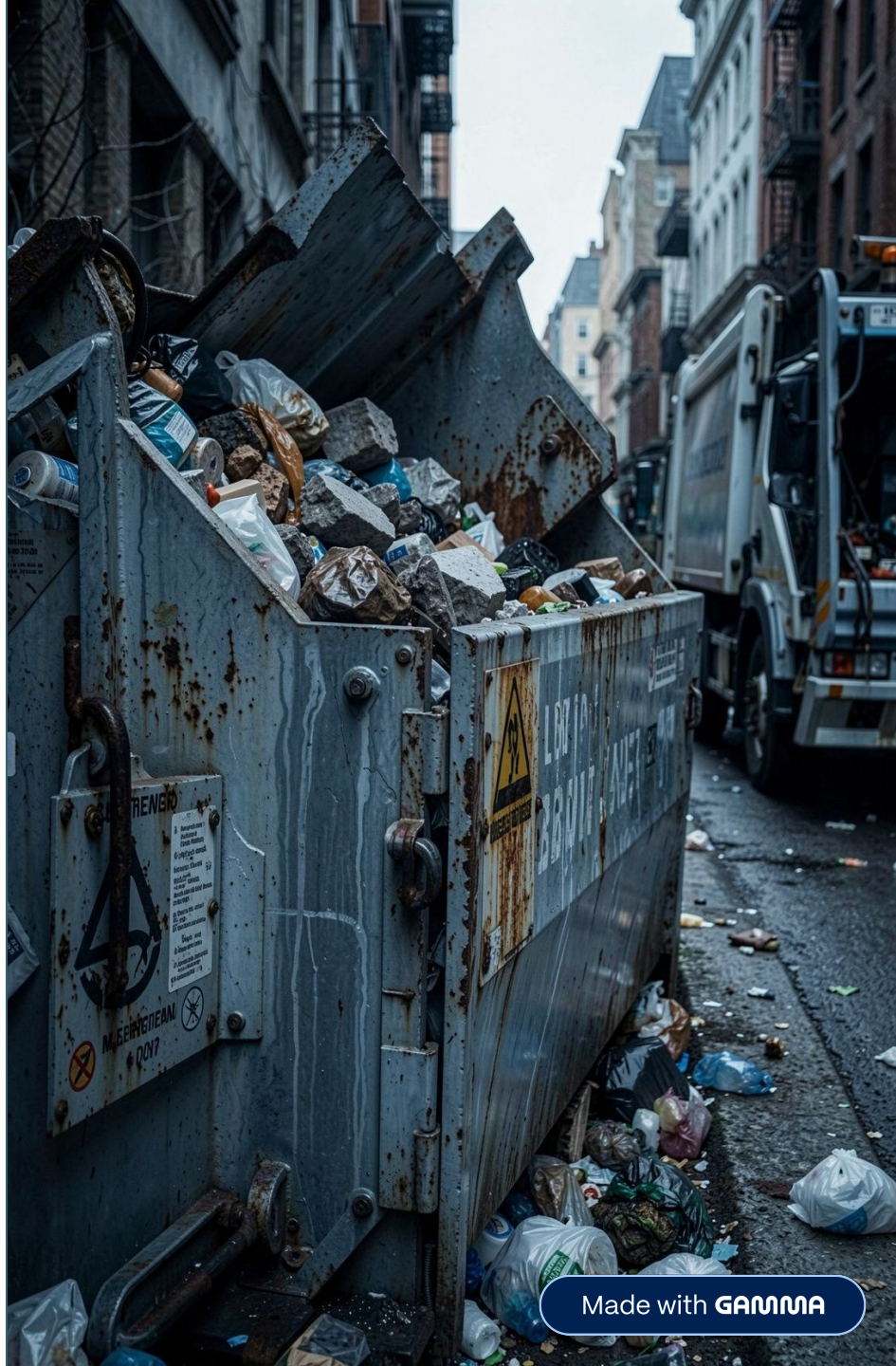
THE OPERATING PROBLEM

Waste is growing faster than cities can manage it

Population growth, urbanization, and consumption are straining collection systems and budgets, while operational decisions remain largely reactive.

When signals arrive late, a collection route becomes an emergency response instead of a planned decision.

الشرح: الضغوط السكانية والتحضر تُثقل أنظمة الجمع والميزانيات، والقرارات ما زالت تفاعلية.



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FROM RESPONSE TO PREDICTION

The answer is not more bins, but smarter decisions

Sensing, analytics, and action operate as one closed operating loop.

Continuous sensing

Sensors track fill level, weight, and temperature.

Actionable prediction

AI forecasts fill levels and recommends routes and timing.

Unified control

One dashboard links the city, drivers, and sorting facilities.

الشرح: تربط المنظومة الاستشعار والتحليل والتنفيذ في حلقة واحدة تقلل الاستجابة المتأخرة.

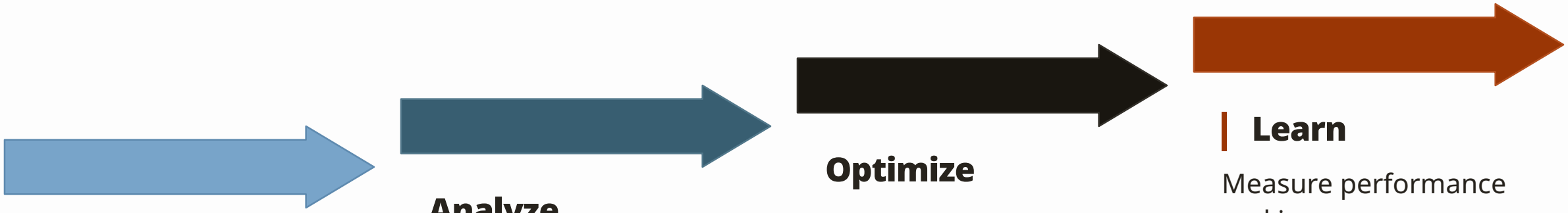


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DECISION CHAIN

From bin data to decisions in four steps

Raw signals become measurable operational action.



Collect

Gather data from bins, vehicles, and cameras.

Analyze

Detect patterns and forecast fill levels and risks.

Optimize

Optimize routes and schedules in real time.

Learn

Measure performance and improve continuously.

الشرح: تتحول البيانات الخام عبر أربع خطوات إلى إجراء قابل للقياس.

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COMPUTER VISION

Computer vision raises sorting value



Computer vision analyzes camera images to identify materials. Cameras separate recyclables from contaminants, improving material quality and reducing reliance on manual sorting alone.

> Clearer detection at the sorting point means cleaner material downstream.

الشرح: الكاميرات تفصل المواد القابلة لإعادة التدوير عن الملوثات، فترتفع جودة المادة.

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FLEET & FIELD TEAMS

Smart fleet operations cut waste and improve safety



Route visibility

Track vehicles and monitor trip status.



Early alerts

Warnings for hazards, blind spots, and unsafe driving.



Measured improvement

Operational data shows efficiency gains and fewer incidents.

الشرح: تتبع المركبات وتنبيهات السلامة والبيانات التشغيلية تقلل الهدر وتحسن الأمان.

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SHARED VALUE

Value appears at three levels

Every better decision improves service, operations, and the environment.



City

Cleaner service and faster response.



Operator

More precise routes and less time on unnecessary trips.



Environment

More waste diverted from landfill and higher recycling quality.

الشرح: تتحقق الفائدة للمدينة والمشغّل والبيئة معًا عند تحسين القرار.

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GETTING STARTED

Start with a small, measurable pilot

Choose one clear problem, then scale what proves its impact.



Define

Pick a district or industrial zone with a clear problem.

Connect

Install sensors and connect vehicle data.

Test

Test fill-level forecasting and route optimization.

Scale

Compare before and after, then scale what works.

الشرح: ابدأ بموقع واحد، واربط البيانات، وقِس النتائج قبل التوسع.

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SUCCESS FACTORS

Success requires technology and governance together



Trusted data

Reliable data and agreed performance indicators.



Clear privacy

Transparent privacy rules for cameras and location data.



Team capability

Train operations teams rather than replace their expertise.



System integration

Integration with sorting and recycling facilities.

الشرح: نجاح المشروع يعتمد على بيانات موثوقة، وخصوصية واضحة، وتدريب الفرق، وتكامل مع مرافق الفرز.

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SMART WASTE AI TURNING WASTE INTO MANAGEABLE FLOW

Next step: pick a pilot zone, capture a baseline, and test one AI-supported decision.

From field signal to measurable operational decision

الشرح: الخطوة التالية هي اختيار منطقة تجريبية، وقياس خط الأساس، واختبار قرار واحد يمكن للذكاء الاصطناعي تحسينه.

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